NOTRE DAME UNIVERSITY

BANGLADESH

EEE Lab Report-06

Course Title: Electrical Circuit Analysis Lab

Course Code: EEE-2108

Experiment No: 06

Name of the Experiment: AC to DC Conversion & measurement of ripple voltage by frequency variation.

Date of Experiment: 24-08-23

Date of Submission: 14-09-23

Submitted by:

Group: E

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Batch: CSE-20

Submitted To:

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Lecturer, NDUB

Experiment No. 06

Name of Experiment: AC to DC conversion and measurement of ripple voltage by frequency variation.

Theory: AC or Alternating Current is that, which is time varing current or voltage that changes the flow direction from time to time. In an ac signal after stepping down of voltage, AC is converted to DC using rectifiers, such as diodes. A full bridge rectifier can be used to convert AC to DC. In the other hand setting up an resistance with a capacitor parallely with a diode in series, we can construct a circuit to convert AC signal to DC. At this case after rectification, the signal is still pulsing, and to make it smoother, a capacitor is used as a filter element. And this capacitor creates the ripple ~~volta~~ voltage in the output signal.

There are 2 ways to decrease this ripple voltage & gain the waveshapes more smooth. They are –

i) By increasing the capacity of the capacitor.

ii) By increasing the value of inputed frequency.

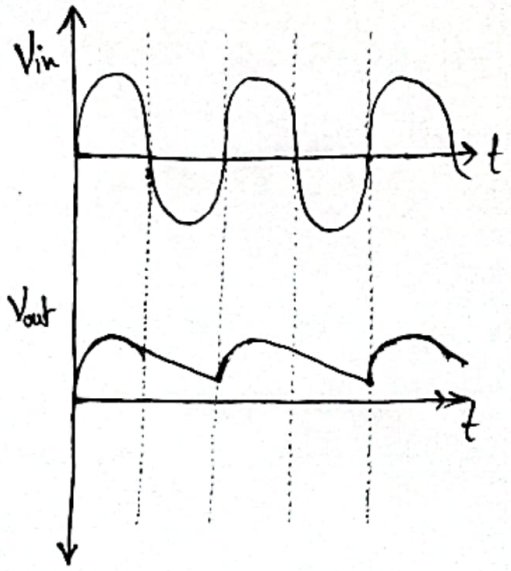


Fig: AC Signal to DC Signal

Equipments & Components:

1. Diode – 1 piece
2. Resistor – 10kΩ - 1 piece
3. Trainer board
4. Capacitor - 0·1 μF - 1 piece
5. AC Signal generator
6. Digital Storage Oscilloscope
7. Chords & wire.

Procedures:

1. Connections are made as per the ~~circuit~~ circuit diagram from 6.1 using conventional diode.

2. Switch the AC Signal generator. Vary input frequency 500Hz, 1000Hz & 5000Hz.

3. Now connect to circuit according to the fig 6.1 & Set the Digital Storage Oscilloscope with the circuit.

4. For various frequency of AC inputed signal, we will get the waveshapes signal same as fig 6.2, 6.3 ~~& 6.4~~.

5. Now calculating the ripple voltage of every individual frequency of the circuit using the formual -

$$\boxed{V_r = \frac{V_p}{fRC}}$$

& fill the table.

6. We will be able to see the DC Signal in the Digital Storage Oscilloscope.

AC to DC Conversion Circuit & Wave shapes:

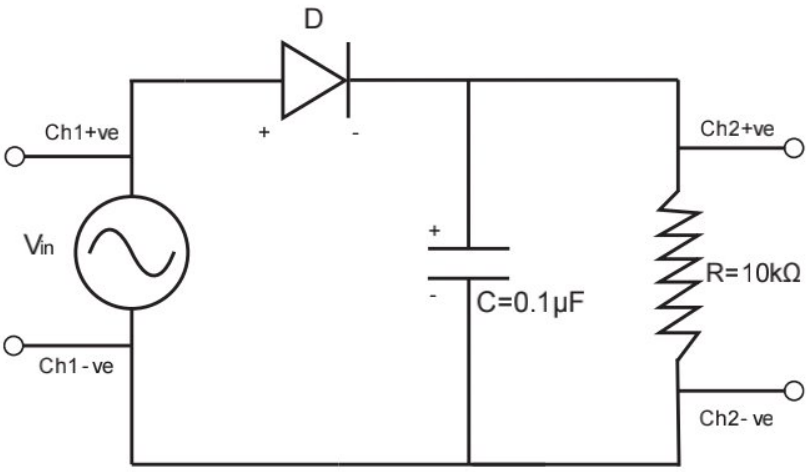


Fig 6.1: AC to DC signal Conversion Circuit

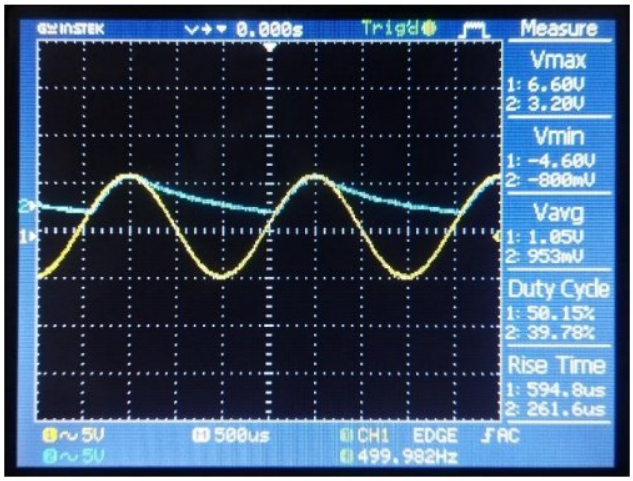
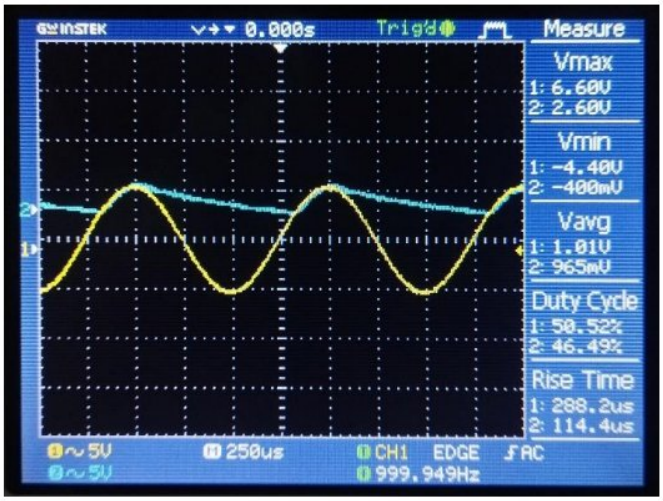


Fig 6.2: Output Wave shape (Frequency 500Hz)



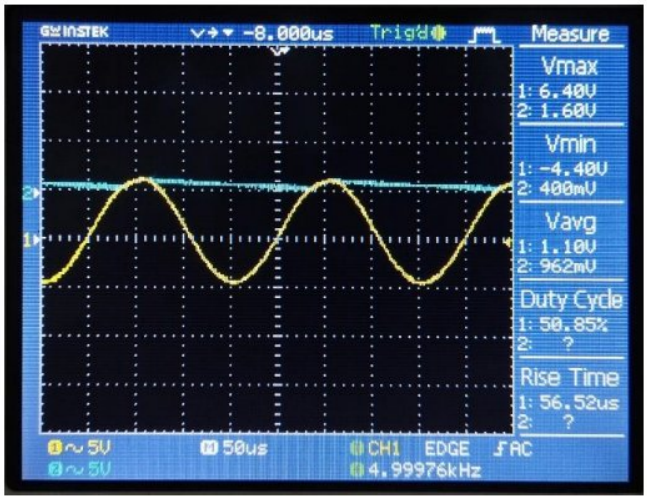


Fig 6.3: Oscilloscope Output Wave Shape for frequency 1KHz & 5KHz

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Experiment-6:- AC to DC conversion and measurement of ripple voltage by frequency variation

	Frequency (f)	V_r (V)
1	500Hz	3.2
2	1000 Hz	2.6
3	5000 Hz	1.6

24/08/23

Student ID: 2, 5, 9, 14

Group No: E

DATA SHEET:

Table: Measuring Ripple Voltage by frequency variation.

Serial no.	Frequency (f) (Hz)	Ripple Voltage V_r (v)
1.	500 Hz	3.2 v
2.	1000 Hz	2.6 v
3.	5000 Hz	1.6 v

Discussion :

1. Calibrate the oscilloscope for accuracy.
2. Use low voltage & current.
3. Be contious with reverse voltage to prevent damage.
4. Measure voltage in parallel & when measuring current, short the circuit.
5. Avoid overloading & short circuit.
6. Follow guidelines & manual.